



Why is ToRadicals[] not able to handle all cases? Is there a workaround?

Asked 12 years, 11 months ago Modified 12 years, 11 months ago Viewed 855 times



The documentation for the function [ToRadicals](#) says:

14

◆ There are some cases in which expressions involving radicals can in principle be given, but `ToRadicals` cannot find them.



I'm concerned only with cases where the argument to `ToRadicals` is a single `Root` object with a single polynomial function with integer coefficients and no parameters, and an explicit root index.



Here is an example when *Mathematica* cannot find an expression in radicals:

```
ToRadicals[Root[-1 - #1^2 - #1^3 + #1^4 + #1^6 &, 2]]
```

But actually the expression exists:

$$\frac{1}{\sqrt[3]{36}} \left(\frac{3}{\sqrt{\beta}} + \frac{\beta}{2} \right), \text{ where } \beta = \sqrt[3]{2}\alpha - 8\sqrt[3]{\frac{3}{\alpha}}, \alpha = 9 + \sqrt{849}.$$

Another example is

```
ToRadicals[Root[6 + 25 #1 - 25 #1^3 + 5 #1^5 &, 5]]
```

where the expression in radicals is

$$\sqrt[5]{\frac{-3 + 4\sqrt{-1}}{5}} + \sqrt[5]{\frac{-3 - 4\sqrt{-1}}{5}}$$

What is the nature of this restriction? Is the problem known to be undecidable in general? Or is it extremely computationally expensive?

Can we write an implementation of `ToRadicals` that is able to find an expression in radicals in all cases when it is possible? Or, at least, in much more cases than the built-in `ToRadicals` does?

equation-solving

functions

symbolic

algebraic-manipulation

expression-manipulation

Share Edit Close Delete Flag

edited Oct 14, 2013 at 20:51

asked Oct 14, 2013 at 18:43



Vladimir Reshetnikov

6,651 32 83

- 3 Requires computational Galois theory in general. It's computationally expensive and probably nowhere implemented in Mathematica. – Daniel Lichtblau Oct 14, 2013 at 19:24

2 Answers

Sorted by: Reset to default Date modified (newest first)



5



```
rad = Root[-1 - #^2 - #^3 + #^4 + #^6 &, 2];
poly = rad[[1]][x]
PolynomialRemainder[poly, x^2 + a x + b, x]
a /. SolveAlways[% == 0, x];
Factor[poly, Extension -> %[[1]]];
x /. Solve[% == 0, x];
ans = Select[%, N[# == rad] &] // First // Simplify
```

$-1 - x^2 - x^3 + x^4 + x^6$

$-1 + b - a b - a^2 b - a^4 b + b^2 + 3 a^2 b^2 - b^3 + (a - a^2 - a^3 - a^5 + b + 2 a b + 4 a^3 b - 3 a b^2) x$

$\frac{1}{12} (8 3^{(2/3)} (2/(-9 + \text{Sqrt}[849]))^{(1/3)} - 2^{(2/3)} (3 (-9 + \text{Sqrt}[849]))^{(1/3)} + \text{Sqrt}[2 (24 + 96 3^{(1/3)} (2/(-9 + \text{Sqrt}[849]))^{(2/3)} + 2^{(1/3)} (3 (-9 + \text{Sqrt}[849]))^{(2/3)})])$

Share Edit Follow Flag

edited Oct 14, 2013 at 19:12

answered Oct 14, 2013 at 18:58



chyanog

16.3k 3 47 89

This approach works on my original example, but fails to express `Root[6 + 25 # - 25 #^3 + 5 #^5 &, 5]` in radicals. But actually it is $\sqrt[5]{\frac{-3+4\sqrt{-1}}{5}} + \sqrt[5]{\frac{-3-4\sqrt{-1}}{5}}$. – Vladimir Reshetnikov Oct 14, 2013 at 20:38



4



We can solve your quintic by substituting $x = y + \frac{1}{y}$. This gives the equation $5y^5 + 6 + \frac{5}{y^5} = 0$, which is quadratic in y^5 .

```
In[1]:= Module[{poly, y, sol},
  poly = 6 + 25x - 25x^3 + 5x^5;
  sol = y + 1/y /. Solve[0 == poly /. x -> y+1/y, y];
  Union[Select[sol, Chop[N[poly /. x -> #]] == 0&], SameTest -> (N[#1] ==
N[#2]&)]
]

Out[1]= {
  1/(-(3/5)-(4 I)/5)^(1/5)+(-(3/5)-(4 I)/5)^(1/5),
  (-1)^(4/5)/(-(3/5)-(4 I)/5)^(1/5)-(-1)^(1/5) (-(3/5)-(4 I)/5)^(1/5),
  -((-1)^(3/5)/(-(3/5)-(4 I)/5)^(1/5))+(-1)^(2/5) (-(3/5)-(4 I)/5)^(1/5),
  (-1)^(2/5)/(-(3/5)-(4 I)/5)^(1/5)-(-1)^(3/5) (-(3/5)-(4 I)/5)^(1/5),
  -((-5)^(1/5)/(-3-4 I)^(1/5))+(-1)^(4/5) (-(3/5)-(4 I)/5)^(1/5)
}
```

I got lucky here trying to sub $x = y + \frac{a}{y}$. Here's *Mathematica* code, written by Michael Trott, to solve any quintic solvable in radicals. It's old code, so some modifications will need to be made. *It would be really cool if this was incorporated into ToRadicals!*

<http://library.wolfram.com/infocenter/Demos/158/>

Unfortunately, I'm unable to get it to solve your quintic though!

Edit

I think *ToRadicals* is generally intended for cubics and quartics. Trying to find roots of simple quintics such as $(x + 1)^5 + 2$ in terms of radicals doesn't even seem to work out.

```
In[2]:= Solve[3 + 5 x + 10 x^2 + 10 x^3 + 5 x^4 + x^5 == 0, x] // ToRadicals

Out[2]= {
  {x -> Root[3 + 5 #1 + 10 #1^2 + 10 #1^3 + 5 #1^4 + #1^5 &, 1]},
  {x -> Root[3 + 5 #1 + 10 #1^2 + 10 #1^3 + 5 #1^4 + #1^5 &, 2]},
  {x -> Root[3 + 5 #1 + 10 #1^2 + 10 #1^3 + 5 #1^4 + #1^5 &, 3]},
  {x -> Root[3 + 5 #1 + 10 #1^2 + 10 #1^3 + 5 #1^4 + #1^5 &, 4]},
  {x -> Root[3 + 5 #1 + 10 #1^2 + 10 #1^3 + 5 #1^4 + #1^5 &, 5]}
}
```

Share Edit Follow Flag

edited Oct 14, 2013 at 21:48

answered Oct 14, 2013 at 21:38



Greg Hurst

39.7k

1

103

149